

# Anticipatory Behavioural Responses to Environmental Policy Announcements: Evidence from the London Ultra Low Emission Zone

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2026-08-01

**Abstract**

**ABSTRACT**

# 1 Introduction

Standard models of regulatory compliance assume behavioural change occurs at the point of implementation. Yet if economic agents are forward-looking and form rational expectations, as Muth (1961) first formalised, credible future policy may alter present behaviour well before implementation begins. Kydland and Prescott (1977) demonstrated that the effectiveness of policy depends fundamentally on its credibility: agents who believe a policy will be implemented adjust accordingly, while announcements perceived as reversible or discretionary will be ignored. Lucas (1976) showed more generally that behavioural models are not invariant to policy regime, since the decision rules being estimated already embed agents' expectations about future policy. Applied to the gap between announcement and enforcement, this implies that treating behaviour as static until formal implementation misspecifies the underlying decision problem. Credible policy announcements may therefore constitute a policy instrument in their own right.

The London Ultra Low Emission Zone (ULEZ) provides a natural empirical setting for studying anticipatory compliance. Introduced in April 2019 under Mayor Sadiq Khan, the scheme charges non-compliant vehicles a daily fee for entering the zone, with the stated aim of reducing the air pollution damaging residents' health and the environment. Two features make it a useful case. First, each expansion was announced well before it took effect. The London-wide expansion of August 2023, for instance, was proposed in March 2022 and confirmed that November, leaving a gap of roughly nine months between the November confirmation and enforcement. That gap provides a window in which anticipatory compliance, if it occurs, can be detected. Second, the zone expanded three times between 2019 and 2023, each announced in advance of implementation. Repeated expansions allow the anticipatory response to be tested across successive waves, and they raise a further question: whether the effect weakens as each new announcement conveys less genuinely new information than the last.

The paper addresses two questions. First, did each ULEZ expansion generate anticipatory compliance before implementation? Second, does the anticipatory effect vary across the three expansions? Both follow from the theoretical claim at the centre of this study: that a credible announcement functions as a policy instrument in its own right, capable of shifting behaviour before enforcement. If this holds, compliance should rise measurably in the window between a credible announcement and its implementation. Whether it does, and whether the response weakens as successive expansions convey less new information, is what the analysis sets out to test.

The analysis uses synthetic control as its primary method. For each treated borough, the method builds a counterfactual from a weighted combination of untreated units, which avoids reliance on the parallel trends assumption required by conventional difference-in-differences. The donor pool for each expansion draws on units not yet treated in that wave: for the 2019 and 2021 expansions this includes outer London boroughs alongside the surrounding districts, while for the 2023 London-wide expansion, where every London borough is eventually treated, the donor pool rests on the surrounding never-treated districts alone. The dataset covers 52 units: 33 London boroughs and 19 surrounding districts serving as never-treated controls.

These form a balanced quarterly panel running from 2011 Q4 to 2024 Q4. The outcome is the ultra low emission vehicle (ULEV) share of licensed cars, constructed from DVLA vehicle licensing records. The synthetic control results are tested against two-way fixed effects difference-in-differences, an event study, and interrupted time series specifications, with a geographic regression discontinuity design considered but limited by the small number of boroughs near the zone boundary.

The analysis finds anticipatory compliance in specific boroughs rather than spread evenly across London, and the effects are consistently modest. For the 2019 expansion, Tower Hamlets shows the strongest evidence of a response, with its ULEV share rising above its synthetic counterfactual in the quarters after the announcement, though the effect is marginal rather than conventionally significant. The 2021 expansion yields no significant effects, with only Hammersmith and Fulham reaching the suggestive range. For the 2023 London-wide expansion, two outer boroughs show significant anticipatory effects once surrounding never-treated districts are used to build the counterfactual: Hillingdon and Sutton.

This study makes three contributions. Theoretically, it extends the literature on anticipatory behaviour, developed largely outside environmental regulation, to a domain where compliance requires a costly and durable purchase. Empirically, it tests this mechanism across three repeated waves of the same policy, where existing work has largely examined single policy events, tracing the anticipatory response as the policy moves from novel to familiar. Methodologically, it uses districts bordering Greater London as never-treated controls, a donor pool that shares the commuter exposure of outer London boroughs without being subject to the zone itself, providing a cleaner counterfactual than inner London boroughs that were themselves treated in earlier waves.

## ROADMAP

## 2 Policy Background

The Ultra Low Emission Zone (ULEZ) is a charging scheme run by Transport for London (TfL) under the Mayor of London. Vehicles that do not meet specified exhaust emission standards pay a daily fee of £12.50 to drive within the zone, applied at all hours. Petrol cars and vans must meet the Euro 4 standard, diesel cars and vans the Euro 6 standard, and motorcycles and mopeds the Euro 3 standard. Compliance is defined by these emission standards rather than by fuel type: a Euro 4 petrol or Euro 6 diesel car is exempt from the charge, as are fully electric and most plug-in hybrid vehicles. Ultra low emission vehicle adoption is therefore one route to compliance among several, but a distinctive one, since it represents the durable, forward-looking capital decision that anticipatory behaviour would predict. The policy aims to improve air quality by lowering concentrations of harmful pollutants, particularly nitrogen dioxide and particulate matter, and to protect public health across London.

ULEZ emerged as a gradual evolution of London’s clean air policy, building on the 2003 Congestion Charge and the 2008 Low Emission Zone that targeted heavy vehicles. It was proposed to address illegal levels of air pollution, it was accelerated and launched in central London by Mayor Sadiq Khan in 2019, driven by rising public health concerns and a series

of legal defeats over the UK’s failure to meet air quality obligations. These earlier schemes meant London drivers were already familiar with zone-based charging, and the enforcement infrastructure of cameras and Automatic Number Plate Recognition (ANPR) was already in place. That existing apparatus made subsequent ULEZ announcements credible: a commitment to expand the zone was believable precisely because the equipment to enforce it already existed.

Launched on 8 April 2019, London’s initial 24-hour ULEZ was mapped to the central Congestion Charge Zone, covering roughly 21 square kilometres and encompassing the high-density commercial core. The scheme was first proposed in 2015 under Mayor Boris Johnson, but the decisive signal came on 4 April 2017, when Mayor Sadiq Khan announced the April 2019 launch date and brought the timetable forward. The same 2017 announcement also flagged a planned extension to the North and South Circular roads, which was formally confirmed on 8 June 2018 with an implementation date of 25 October 2021. This April 2017 announcement is the point from which anticipatory responses to the central zone are measured in the analysis that follows.

The inner expansion to the North and South Circular roads was confirmed in June 2018 and implemented on 25 October 2021. Charges and emission standards remained identical to the central zone, but the scale increased sharply, covering an area roughly eighteen times larger and bringing approximately 3.8 million residents within the zone [Mayor of London Press release]. The gap of over three years between confirmation and enforcement is the longest of the three expansions, giving the widest window in which anticipatory compliance could occur.

The scheme was then proposed for a London-wide expansion in March 2022, confirmed in November 2022, and implemented on 29 August 2023, leaving a nine-month window between confirmation and enforcement. This was the largest expansion, bringing all 32 boroughs and the City of London within the zone (Figure 1). Its credibility was tested directly: a coalition of five councils (four outer London boroughs - Bexley, Bromley, Harrow, and Hillingdon, and Surrey County Council) challenged the expansion in the High Court, but the challenge failed in July 2023 and enforcement proceeded as announced. That the commitment held despite legal challenge reinforced the signal to drivers that the deadline was firm.

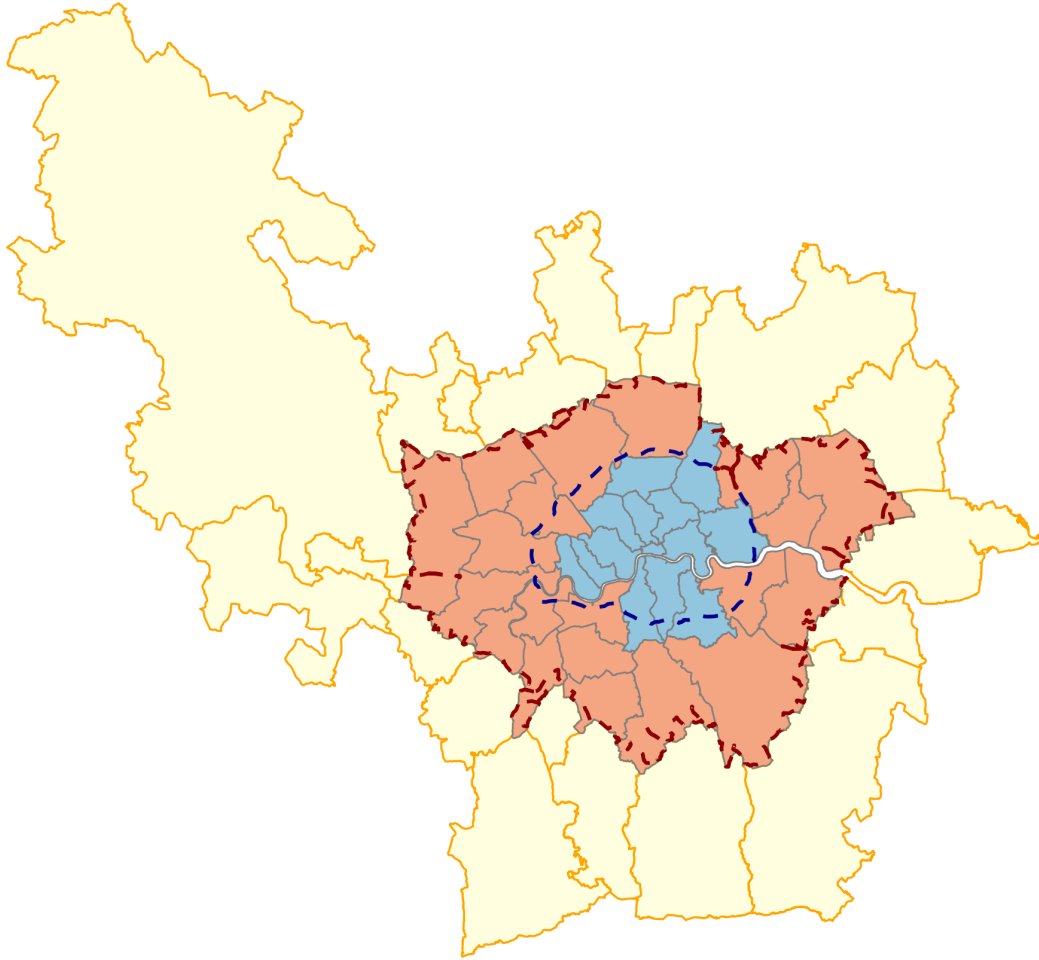


Figure 1: Study area and treatment assignment. Inner London boroughs (blue) were covered by the 2019 and 2021 expansions; outer London boroughs (red) by the 2023 London-wide expansion; the nineteen surrounding districts (yellow) were never subject to the charge. Dashed lines mark the 2021 and 2023 zone boundaries.

The credibility of these commitments rested on three reinforcing foundations. Each expansion was preceded by statutory public consultation and formal confirmation, signaling that the policy followed a legally prescribed procedure rather than an arbitrary mayoral decision. The commitment to charge non-compliant vehicles was underpinned by the camera and ANPR infrastructure already in place, so the announced restrictions could be monitored and penalties imposed from the first day of enforcement. Most decisive was demonstrated follow-through: the 2019 and 2021 expansions were both delivered on their announced timetables, so by the time the 2023 expansion was confirmed, drivers had twice seen the Mayor commit to a future extension and carry it out. Announcements that might once have been read as provisional intentions had become commitments road users could expect to be honoured. Credible, pre-announced expansions created precisely the conditions under which forward-looking

individuals would adjust their behaviour before enforcement began.

### 3 Literature Review

This literature review establishes the foundations for the dissertation in three stages. It first develops the theoretical framework, which explains why credible policy announcements should induce anticipatory behavioural responses before implementation. It then reviews the existing empirical evidence on anticipatory behaviour and low-emission zone policies, assessing what is already known and where important questions remain unanswered. Finally, it turns to the methodological literature, outlining the empirical strategies capable of identifying such anticipation and motivating the research design adopted here. The review concludes by setting out how the research design responds to that gap.

#### 3.1 Theoretical Framework

The theoretical foundation for anticipatory behavioural responses begins with the premise that economic agents make decisions using information about both current and expected future conditions. John F. Muth's rational expectations hypothesis provides the micro-foundation for this argument. Muth (1961) proposed that agents' expectations of future economic variables are not systematically biased but instead coincide, on average, with the predictions implied by the relevant economic model, conditional on the information available to them. Expectations formed this way efficiently incorporate all available information, including credible signals about future policy, rather than relying solely on past experience or adjusting only once change occurs. Although Muth developed this framework in the context of commodity price expectations, its logic extends naturally to announced regulatory costs. When policymakers communicate a future increase in the cost of owning or operating a non-compliant vehicle, that information becomes relevant immediately rather than only when the regulation enters into force. A forward-looking vehicle owner therefore has an incentive to evaluate the future costs of retaining a non-compliant vehicle against the costs of replacing or scrapping it before enforcement begins. Under rational expectations, disregarding information already revealed would itself be irrational, even though the timing of any individual response depends on the costs of adjustment.

This forward-looking behaviour has important implications for how policy effects should be understood empirically. If agents incorporate anticipated policy into present decisions, then observed behaviour before implementation may already reflect the future policy environment. Lucas (1976) formalised this insight through the Lucas critique, arguing that the estimated relationships used to evaluate policy cannot be assumed to remain invariant across policy regimes, because those relationships already embed agents' expectations about the policy environment. Once a future policy is announced, agents revise their decision rules to reflect the expected future environment. Consequently, the period preceding implementation is not necessarily an undisturbed baseline against which policy effects can simply be measured. While the Lucas critique is often invoked as a warning against evaluating policy using naive before-and-after comparisons, it is more constructive in the present context. The possibility that behaviour changes before enforcement is not a methodological nuisance to be eliminated but

the phenomenon of substantive interest. Rather than treating pre-implementation adjustment as a source of bias, this paper asks whether the announcement induced measurable changes in vehicle compliance before the ULEZ expansion came into force.

Whether such anticipatory adjustment occurs, however, depends on a further condition. Rational agents will respond to announced future policy only if they believe that the policy will in fact be implemented as announced. Kydland and Prescott (1977) demonstrate that expectations depend not only on policymakers' announcements but also on the credibility of those announcements. If agents expect a policy to be delayed, reversed, or weakened before implementation, the future regulatory cost remains uncertain and there is little reason to incur the costs of adjustment in advance. Conversely, where a policy commitment is regarded as credible and durable, the expected future cost becomes sufficiently certain to influence present behaviour. Credibility therefore provides the mechanism linking announcements to anticipatory responses. This insight is particularly important for understanding differences across successive ULEZ expansions. Variation in anticipatory compliance across waves need not imply that rational expectations fail; rather, it may reflect differences in the informational content or perceived credibility of successive announcements. As credible commitments accumulate over time, later announcements may convey less genuinely new information, reducing the scope of additional anticipatory adjustment even when the policy itself remains fully credible.

Taken together, these three contributions generate a coherent theoretical framework for analysing anticipatory behavioural responses. Rational agents incorporate credible information about future policy into current decision-making; those expectations imply that behavioural adjustment may begin before implementation, meaning that the pre-implementation period cannot automatically be treated as an unaffected baseline; and such adjustment occurs only when policy announcements are regarded as credible. The resulting prediction is therefore conditional rather than universal: where a credible announcement conveys precise information about a future regulatory cost, forward-looking agents have an incentive to adjust before implementation when the expected gains from acting early outweigh the costs of adjustment. Anticipatory responses should therefore be expected to be partial and heterogeneous rather than uniform across all individuals or policy settings. Although the components of this theoretical chain have been examined separately across a range of economic contexts, they have not been brought together to study anticipatory responses to geographically bounded environmental regulation in the window before implementation — the intersection this dissertation addresses.

## **3.2 Empirical Evidence**

The theoretical framework established the conditions under which anticipatory responses should occur. The next question is whether such responses have been observed empirically. Existing evidence demonstrates that credible policy announcements can induce behavioural adjustment before implementation across a range of settings, while a separate literature shows that low-emission zone policies alter vehicle compliance. These two strands, however, have largely developed independently.

The closest empirical analogue is provided by Mills and Rüttenauer (2022), who examine behavioural responses to the announcement of mandatory COVID-19 certification. Using a synthetic control design, they show that vaccination rates begin increasing approximately twenty days before certification requirements entered into force, demonstrating that a credible announcement of a future access restriction altered behaviour before legal enforcement commenced. Their analysis explicitly backdates the intervention period to capture anticipation rather than treating implementation as the sole point of policy exposure, making it methodologically as well as conceptually similar to this paper. The principal distinction lies in the nature of the treatment. Whereas COVID certification constituted a national policy applying uniformly across the population, the ULEZ expansion is geographically bounded, leaving areas outside the zone untreated while subject to the same regional conditions. This clear separation between treated and untreated areas provides a sharper counterfactual than is available in a nationwide intervention. More broadly, Mills and Rüttenauer establish that anticipatory responses are observable in the context of public health policy, demonstrating the phenomenon outside environmental regulation while leaving open the question of whether similar behaviour arises in geographically targeted environmental policies.

Malani and Reif (2015) provide the conceptual framework for interpreting such anticipatory responses. Studying tort reform across US states, they argue that apparent pre-treatment trends in difference-in-differences designs need not indicate a violation of the parallel trends assumption but may instead reflect genuine behavioural responses to announced policy changes. This distinction is particularly important for this paper, where an increase in ULEZ-compliant vehicles before implementation could be interpreted either as an underlying trend or as anticipation. Their framework supplies the econometric language for adjudicating between these competing explanations, rather than presuming that pre-implementation movement is necessarily confounding. Their contribution also foreshadows the methodological discussion that follows by illustrating how empirical designs must explicitly account for anticipation when estimating policy effects.

Complementary evidence comes from Ito and Sallee (2018), who show that Japanese vehicle manufacturers strategically adjusted vehicle weight in anticipation of fuel-economy regulatory thresholds, building on the bunching framework developed by Saez (2010). Although these studies examine supply-side responses to regulatory notches rather than household vehicle replacement decisions, they reinforce the broader principle that known future policy thresholds induce strategic adjustment before they become binding. The underlying logic is therefore shared, even though the behavioural margin and institutional setting differ substantially from geographically bounded emission-zone policy.

A separate body of literature examines how low-emission zone policies influence vehicle fleet composition after implementation. Ellison, Greaves, and Hensher (2013) provide the foundational analysis of London’s 2008 Low Emission Zone, documenting changes in heavy-vehicle fleet composition alongside modest reductions in particulate emissions using before-and-after comparisons. While their analysis is descriptive and does not employ a formal counterfactual, it establishes that DVLA vehicle licensing data capture meaningful fleet responses to emission-zone policy, providing empirical support for the outcome measure used in this dissertation.



The study most closely aligned with the present setting is Wolff (2014), who investigates Germany’s Low Emission Zone programme and finds that vehicle owners living closer to an LEZ boundary were more likely to register cleaner vehicles, revealing a spatial gradient in compliance driven by proximity to the regulated area. This demonstrates that geographically defined environmental policies can generate measurable changes in vehicle registration behaviour. However, Wolff analyses compliance after the policy is already in force, whereas the present dissertation focuses on the anticipatory period between announcement and implementation. The distinction is important: existing evidence establishes that geography shapes compliance once regulation exists, but not whether geographically bounded announcements induce behavioural adjustment before enforcement begins.

Zhai and Wolff (2021) strengthen this literature through an instrumental variables analysis of London’s Low Emission Zone, showing that compliance responses differed across implementation phases, with short-run adjustments diverging from longer-run environmental improvements. Their findings demonstrate that fleet responses evolve over time rather than following a simple monotonic path, providing important context for interpreting behavioural adjustment around successive policy expansions.

Taken together, these empirical literatures establish two complementary findings. First, credible policy announcements can induce anticipatory behavioural responses across a variety of policy domains. Second, geographically defined emission-zone policies produce measurable changes in vehicle compliance once implemented. However, no existing study brings these strands together by examining whether announcements of geographically bounded emission-zone expansions generate measurable compliance responses before implementation. Addressing that question requires an empirical strategy capable of identifying anticipation while constructing a credible counterfactual, to which the methodological literature reviewed below now turns.

### **3.3 Methodological Literature**

The preceding section implied that identifying anticipatory behaviour requires an empirical design capable of satisfying two conditions: constructing a credible counterfactual without assuming an undisturbed pre-treatment period, and ensuring that comparison units remain genuinely unaffected by the policy announcement. Because policy anticipation may emerge before implementation, the appropriate methodological question is not simply whether outcomes change after enforcement, but which empirical strategies can credibly isolate behavioural responses to the announcement itself while providing valid inference with a small number of treated units.

The synthetic control method provides the principal framework for addressing these requirements. Introduced by Abadie and Gardeazabal (2003) and subsequently formalised by Abadie, Diamond, and Hainmueller (2010), the approach estimates the counterfactual outcome for a treated unit by constructing a weighted combination of untreated units that closely reproduces its pre-treatment trajectory. Rather than assuming that treated and untreated units would have followed parallel trends, the synthetic control estimator uses observed pre-treatment outcomes and predictors to approximate what would have occurred in the absence of treatment.

This feature is particularly well suited to the present setting. As the theoretical discussion of anticipatory behaviour suggests, policy announcements may already influence behaviour before implementation, making the pre-treatment period an unreliable benchmark if treated and untreated units are simply assumed to evolve in parallel. Synthetic control instead constructs that benchmark directly from the data. Abadie, Diamond, and Hainmueller (2015) further develop placebo-based inference, assessing whether the estimated treatment effect is unusually large relative to placebo effects generated by assigning treatment to each donor unit in turn. Statistical significance therefore derives from the treated unit’s position within the placebo distribution rather than from conventional sampling theory. Inference consequently depends on the size of the available donor pool, and placebo diagnostics remain informative alongside reported p-values. Abadie (2021) synthesises this literature and emphasises a further requirement that is central to the present dissertation: donor units must be plausibly unaffected by the intervention itself, while excessively large or heterogeneous donor pools may weaken the credibility of the constructed counterfactual.

The choice of comparison units is further informed by the literature on staggered policy adoption. Goodman-Bacon (2021) demonstrates that two-way fixed effects estimators in staggered treatment settings decompose into multiple two-by-two comparisons, some of which use already-treated units as controls for later-treated units. When treatment effects are heterogeneous or evolve over time, these comparisons can generate biased estimates and even negative weighting. Callaway and Sant’Anna (2021) address this problem by estimating group-time average treatment effects using only never-treated or not-yet-treated units as comparison groups. This framework is directly relevant to the successive ULEZ expansions, where treatment is introduced in multiple waves rather than simultaneously. It provides the methodological justification for constructing donor pools from units that have not yet been exposed to the relevant announcement and explains why conventional two-way fixed effects difference-in-differences serves as a robustness check rather than the primary empirical strategy, alongside complementary event-study and interrupted time-series specifications.

A complementary methodological perspective is provided by the geographic regression discontinuity literature. Keele and Titiunik (2015) establish the conditions under which administrative boundaries can identify causal effects, requiring continuity of potential outcomes at the boundary, the absence of coincident treatments, and sufficient observations close to the discontinuity. Dell’s (2010) study of Peru’s historical mita boundary demonstrates the practical application of this design, showing that administrative borders can reveal persistent causal effects when these identifying assumptions are satisfied. Because the ULEZ expansion is defined by an administrative boundary, a geographic regression discontinuity design is available in principle. However, the limited number of boroughs adjacent to the boundary and the presence of broader administrative differences make these identifying assumptions considerably more demanding than those underpinning the synthetic control approach. Section 6 assesses this design for feasibility rather than adopting it as a robustness check.

Taken together, these methodological contributions establish the principles that guide the empirical strategy adopted in this dissertation. The analysis requires a design that constructs an explicit counterfactual rather than assuming parallel trends, compares treated units only with areas plausibly unaffected by the relevant policy announcement, and supports credible

inference despite a limited number of treated units. Synthetic control therefore serves as the primary research design, implemented using never-treated surrounding districts alongside not-yet-treated London boroughs as donor units, with placebo-based inference. Two-way fixed effects difference-in-differences, event-study and interrupted time-series specifications provide complementary robustness checks, while a geographic regression discontinuity design is assessed for feasibility but cannot be credibly identified at borough level.

## 4 Data and Study Design

### 4.1 Data Sources and Construction

The outcome of interest is the share of licensed cars registered within each local authority that are classified as Ultra Low Emission Vehicles (ULEVs). Vehicle registration data are obtained from the UK Driver and Vehicle Licensing Agency (DVLA) quarterly licensing statistics (Tables VEH0132 and VEH9901). The ULEV classification follows Department for Transport convention, covering vehicles emitting below the prevailing carbon dioxide threshold, principally battery electric and plug-in hybrid cars. The numerator is the number of licensed ULEVs registered within each local authority, while the denominator is the total number of licensed cars registered in the same area. The resulting measure captures the proportion of the local vehicle fleet classified as ULEV. Importantly, this is a stock rather than a flow measure: it reflects the composition of the existing fleet rather than new vehicle registrations. Consequently, any behavioural response to ULEZ announcements is expected to emerge gradually as households replace or retire vehicles over time rather than instantaneously.

As discussed in Section 2, ULEV ownership is not synonymous with ULEZ compliance. Petrol vehicles meeting Euro 4 standards and diesel vehicles meeting Euro 6 standards are equally exempt from the ULEZ charge despite not being classified as ULEVs. The ULEV share is adopted because it captures the form of adjustment that exceeds the compliance threshold rather than merely meeting it. A household that replaces a non-compliant vehicle with a compliant diesel satisfies the announced charge; one that purchases a ULEV commits to a lower-emission vehicle beyond what the regulation requires, a decision more plausibly reflecting expectations about the future direction of regulation than minimal adjustment to a known cost. This is the behavioural margin the theoretical framework predicts. It also has an important implication for interpretation: the estimated effects should be regarded as a lower bound on the total compliance response, since households that replaced non-compliant vehicles with compliant conventional petrol or diesel vehicles are not observed in the outcome measure.

The empirical analysis uses a balanced quarterly panel comprising 52 local authorities observed from 2011 Q4 to 2024 Q4, yielding 2,756 observations. The panel consists of the 33 Greater London local authorities (comprising the 32 London boroughs and the City of London) and 19 surrounding district local authorities. The surrounding districts are selected using a transparent geographic rule. Greater London’s boundary is defined as the union of the 33 borough geometries, and all district authorities with any part of their boundary lying within two kilometres of Greater London are included in the study panel. This criterion identifies

districts immediately adjoining London while avoiding progressively more distant authorities that are less comparable in terms of travel patterns, housing markets, and vehicle ownership. These surrounding districts were never themselves subject to the ULEZ charge, although many of their residents commute into London, making them a plausible source of comparison units throughout the analysis.

In addition to the vehicle licensing data, the panel includes several socio-economic and transport characteristics used in robustness and heterogeneity analyses. These comprise the Index of Multiple Deprivation (IMD), Public Transport Accessibility Levels (PTAL), median house prices, car ownership, population density, and household income. These variables are not used to construct the synthetic control weights themselves, which are based solely on the pre-treatment outcome trajectory, but instead provide additional evidence on the robustness and heterogeneity of the estimated effects.

## 4.2 Treatment Timing and the Dating Rule

The empirical analysis dates each ULEZ expansion to the point at which its implementation became a credible and firm policy commitment rather than to the date on which charging commenced. This approach follows directly from the theoretical framework developed in Section 3. If forward-looking agents respond to credible announcements of future regulatory costs, then behavioural adjustment should begin once implementation becomes certain, not when the policy is implemented. Accordingly, the treatment date for each expansion is defined as the earliest point at which the public could reasonably regard the expansion as a committed policy rather than a proposal subject to revision.

Table 1: Treatment dates by expansion wave

| ULEZ Expansion           | Treatment Date                | Rationale                                                                                                                                                                                                                                                                                   |
|--------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Central London<br>(2019) | 2017 Q2 (4 April<br>2017)     | The central ULEZ had already been confirmed under the previous Mayor in March 2015 for implementation in 2020. The April 2017 announcement brought the implementation date forward to April 2019, altering the timing of an already-committed policy rather than creating a new commitment. |
| Inner London<br>(2021)   | 2018 Q2 (8 June<br>2018)      | The same April 2017 announcement proposed extending the ULEZ to the North and South Circular roads, but implementation remained subject to statutory consultation. The policy became a firm commitment only when the expansion was formally confirmed following that consultation.          |
| Outer London<br>(2023)   | 2022 Q4 (25<br>November 2022) | The March 2022 consultation proposed a London-wide expansion, with implementation contingent on the statutory consultation process. The confirmed decision of 25 November 2022 represents the point at which the expansion became a credible commitment.                                    |

- 5 Results
- 6 Robustness Checks
- 7 Heterogeneity Analysis
- 8 Discussion and Conclusion